INTERNATIONAL CONFERENCE ON PUBLIC ECONOMIC THEORY

Public Sector Financing Needs and the Multi-Year Plan Maturity Index in the Municipalities of Parana

A panel Probit analysis of fiscal adjustment and budget-planning institutions

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MOTIVATION

Fiscal heterogeneity and the limits of adjustment

- Brazilian municipalities are deeply heterogeneous in fiscal capacity and economic dynamism.
- The 1988 Constitution decentralized spending but concentrated revenue — a persistent vertical fiscal gap.
- High budgetary rigidity plus transfer dependence sharply constrain subnational adjustment.
- Sustainability hinges not only on revenue or expenditure levels, but on the quality of fiscal institutions.

Vertical fiscal gap

Revenue concentrated in the Union; spending pushed to municipalities.

Budget rigidity

Payroll and mandatory outlays limit short-run reallocation.

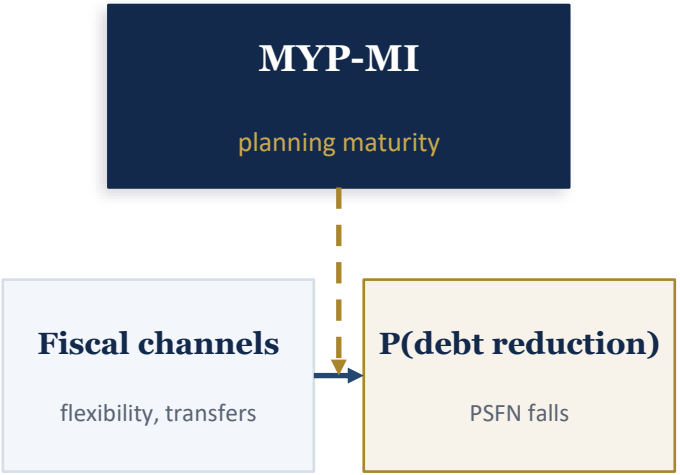
Transfer dependence

Reliance on the MPF can dull local fiscal effort.

RESEARCH QUESTION

Does planning maturity moderate fiscal adjustment?

- Research question: how does budget-planning maturity shape municipal debt dynamics?
- Central hypothesis: the effectiveness of fiscal adjustment is conditioned by the quality of planning institutions.
- Performance-budget planning acts as a moderating factor over fiscal channels.
- **It should amplify expenditure flexibility and attenuate the adverse effects of transfer dependence.**



Second-generation fiscal federalism and institutions

Transfers & incentives

Intergovernmental transfers equalize resources but can induce opportunistic behavior and reduce local fiscal effort.

Oates 2005; Weingast 2009; OECD 2021

Expenditure rigidity

Current spending — payroll, service provision — is institutionally rigid, limiting short-run adjustment.

Dafflon & Vaillancourt 2022

Revenue composition

Volatile bases (services) raise fiscal vulnerability; property taxes add stability and predictability.

Norregaard 2013; OECD 2021

Planning capacity

Planning quality drives intertemporal coordination, yet subnational evidence remains scarce.

Alesina, Favero & Giavazzi 2019

PART I

The municipal budget constraint

From debt dynamics to public sector financing needs (PSFN)

Debt accumulation and its municipal components

Standard intertemporal debt dynamics: debt evolves with the cost of carrying past debt and the primary balance (Blanchard 1990; Bohn 1998).

$$B_{it} = (1 + r_t)\, B_{i,t-1} - T_{it} - G_{it} \qquad (1)$$

For municipalities, revenue splits into own effort and transfers, while expenditure tracks local activity:

$$T_{it} = t^{P}_{it} Y_{it} + t^{T}_{it} Y_t \qquad (2)$$

$$G_{it} = \theta_{it} Y_{it} \qquad (3)$$

Y = economic activity · $t\,P$, $t\,T$ = own-revenue & transfer sensitivities · θ = expenditure rigidity

Recasting in debt-to-revenue terms

Substituting (2)-(3) into (1) and dividing by total revenue (the Fiscal Responsibility Law metric) yields a debt/revenue formulation:

$$B_{it} = (1 + r_t)\, B_{i,t-1} - (t^P_{it} - \theta_{it})\, Y_{it} + t^T_{it} Y_t \qquad (4)$$

Introduce the revenue/GDP ratio t_{it}, revenue growth ρ, and the municipal share of state GDP φ_{it}:

$$\Delta b_{it} = (r_t - \rho)\, b_{i,t-1} - (1/t_{it})\, [\, (t^P_{it} - \theta_{it}) + t^T_{it}\, (1/\varphi_{it})\,] \qquad (6)$$

Public Sector Financing Needs (PSFN)

Equation (7) decomposes the PSFN into two economically distinct components:

$$\Delta b_{it} = (r_t - \rho)\, b_{i,t-1} - (1/t_{it})\,[\,(t^P_{it} - \theta_{it}) + t^T_{it}/\varphi_{it}\,] \qquad (7)$$

Debt coefficient

$(r - \rho) \cdot b$

Interest-rate cost net of revenue growth, applied to the inherited debt stock.

Fiscal effort = primary balance

governed by five parameters: $t, t\,P, \theta, t\,T, \varphi$

The five parameters of fiscal effort

t_{it}	t^{P}_{it}	θ_{it}	t^{T}_{it}	φ_{it}
Tax intensity	**Own-revenue mix**	**Expenditure rigidity**	**Transfer dependence**	**Economic relevance**
Revenue/GDP ratio; a Laffer-type ceiling on effort.	Distortive vs. non-distortive taxes; base elasticity.	Sensitivity of spending to activity; flexibility margin.	Weight of MPF transfers; can reverse the expected sign.	Municipal share of state GDP; own-base capacity.

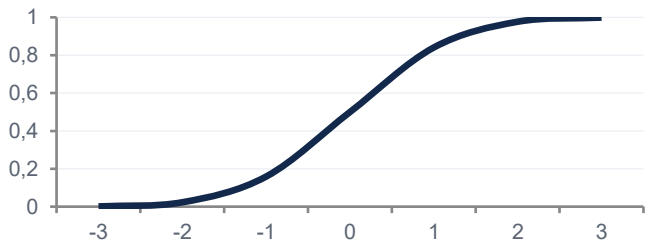
A panel Probit for the probability of adjustment

- Binary outcome: Δb = 1 if the net-debt/revenue ratio falls, 0 otherwise.
- Probit link maps the index to a probability via the standard-normal CDF.
- Nonlinear MLE; effects read through marginal effects, not raw coefficients.
- MYP-MI dummy (>60%) interacted with expenditure flexibility and transfer dependence.
- **Panel: 399 municipalities × 2 MYP cycles = 798 observations.**

$$p = F(\beta_1 + \beta' x') \qquad (14)$$

$$dp/dx' = f(\beta_1 + \beta' x')\,\beta' \qquad (15)$$

Probit link: standard-normal CDF



MEASUREMENT

The Multi-Year Plan Maturity Index (MYP-MI)

- Built by CAS-PR; inspired by the IMF performance-based budgeting model (Robinson & Last, 2010).
- Measures formalized planning quality — not general state capacity.
- Covers all 399 municipalities, longitudinally, with low subjectivity.
- **Dummy = 1 when MYP-MI > 60% ("Developed" and "Improved" strata).**

$$\text{MYP-MI} = (I_1 + I_2 + \cdots + I_7) / 7$$

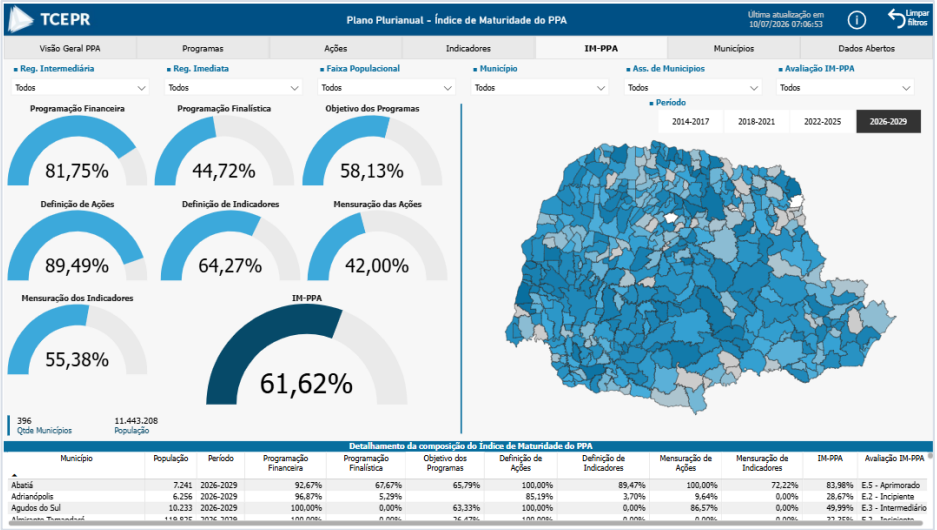
Seven sub-indices (0–1), equally weighted

I_1	Financial Programming
I_2	Core Programs
I_3	Program Objectives
I_4	Action Definition
I_5	Indicator Definition
I_6	Action Measurement
I_7	Indicator Measurement

DATA SOURCE

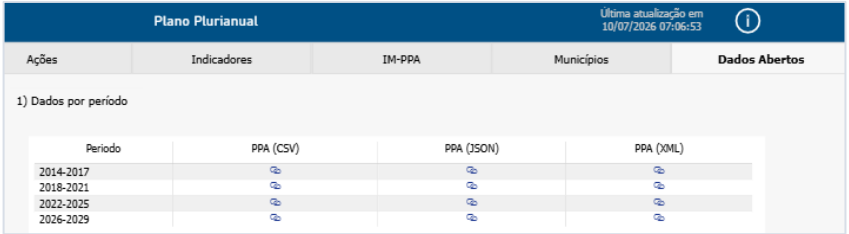
The MYP-MI on the CAS-PR public dashboard

- The MYP-MI is published on an interactive public dashboard maintained by CAS-PR.
- The seven sub-indices, the strata and the composite index can be filtered by municipality and period.
- **Results span all 399 municipalities and are mapped across the state of Parana.**



Open data, freely available to researchers

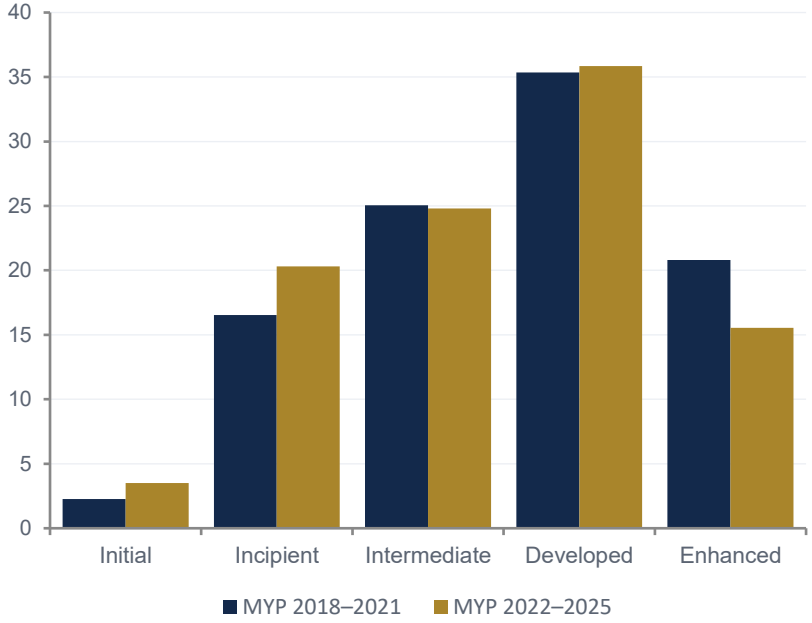
- The data used in this study are open data, published on the Court of Auditors' portal for anyone to download.
- Researchers can retrieve the full series directly, in CSV, JSON and XML formats.
- **Four MYP cycles are now available for analysis: 2014-2017, 2018-2021, 2022-2025 and 2026-2029.**
- The added cycle lengthens the panel and opens the way to dynamic identification strategies.



DATA

Distribution and evolution of planning maturity

- Two complete MYP cycles: 2018–2021 and 2022–2025.
- Fiscal variables averaged within each cycle; deflated to 2025 prices (IPCA).
- 798 observations = 399 municipalities × 2 cycles.
- **Polarization: the “Enhanced” share fell 5.3 p.p.; lower strata grew.**



Share of municipalities by maturity stratum (%)

RESULTS

Panel Probit estimates (marginal effects)

Variable	Marginal effect	z	p-value	Sig.
Current revenue (ln)	**−0.112**	−2.93	**0.003**	***
θ (expenditure flexibility)	0.477	1.35	0.176	n.s.
θ × MYP60	**−1.025**	−2.31	**0.020**	**
t T (transfer dependence)	**−1.011**	−5.04	**0.000**	***
t T × MYP60	**0.480**	2.45	**0.014**	**
t P (own-revenue mix)	**−0.099**	−1.79	**0.073**	*
t P × MYP60	0.029	0.40	0.690	n.s.
t (tax intensity)	0.561	1.51	0.132	n.s.
ϕ (share of state GDP)	**0.092**	1.99	**0.047**	**

Model fit

Observations

798

McFadden R^2

0.039

Correct predictions

62.4%

LR χ^2 (Prob>χ^2)

42.67 (0.000)

**** 1% ** 5% * 10% · n.s. = not significant · bold rows = MYP-MI interactions*

DISCUSSION

Planning maturity as a moderating mechanism

Transfer dependence Higher MPF reliance lowers the probability of adjustment — a moral-hazard channel on fiscal effort.

Moderation of transfers The positive interaction shows maturity attenuates — though does not erase — that adverse effect.

Conditional flexibility Expenditure flexibility matters only jointly with maturity: institutions activate the instrument.

Volatile revenue mix Reliance on service taxes lowers adjustment odds; this channel is not moderated by planning.

Economic relevance Larger municipalities (higher share of state GDP) adjust more readily via their own base.

CONCLUSION

Fiscal sustainability is an institutional outcome

- Subnational sustainability is not determined by economic or financial variables alone.
- MYP maturity is a moderating mechanism: it amplifies flexibility and mitigates transfer dependence.
- Complementarity between fiscal instruments and planning capacity — not a substitute for either.
- Institutional effects are selective: the own-revenue channel is unmoved by planning quality.

Policy implications

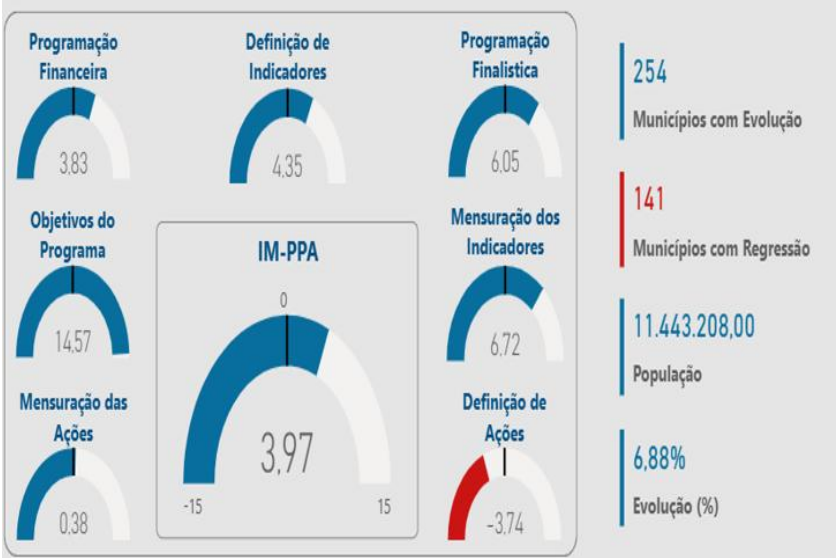
- Strengthen performance-budget planning as a real coordination tool, not a formality.
- Integrate the MYP with the LDO and LOA; define clear goals and indicators.
- Raise transparency of measurement across the planning cycle.
- **Build local state capacity to internalize fiscal constraints.**

Limitations and closing remarks

- Short panel: only two MYP cycles constrain the time dimension.
- Potential endogeneity between fiscal and planning variables is not yet addressed.
- Nonlinear interactions: interpret sign and significance, not raw magnitude (Ai & Norton 2003).
- **Contribution: structural derivation plus evidence of the moderating role of planning.**

Monitoring the maturity indicator's evolution

- Because the maturity index is decisive for improving the fiscal context, CAS-PR tracks its evolution cycle over cycle.
- The new cycle rose by 3.97 percentage points, with 254 municipalities advancing and 141 regressing.
- **A direct bridge between the academic study and its practical application in public management.**



Thank you

Questions and comments are warmly welcome

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